The Enterprise Schematic: Fundamentals of Business IT Architecture

Aligning Strategy, Capabilities, and Technology for Maximum Value.



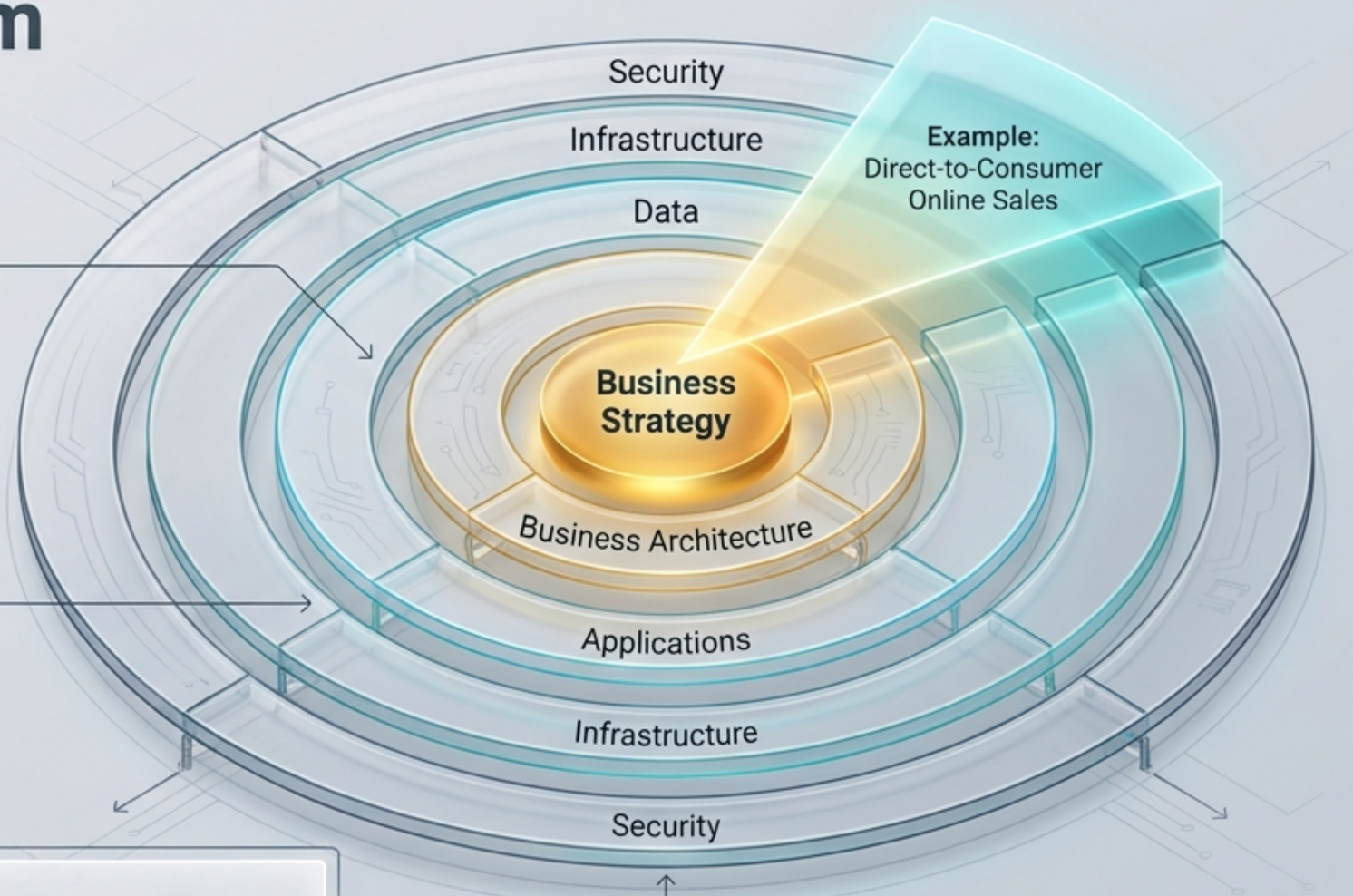
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The Alignment Ripple: Business and IT are One System

Decisions ripple. A new business model forces changes in capabilities, cascading directly into apps, data flows, and infrastructure.

Without an architecture view, projects solve isolated problems, resulting in duplicated systems, inconsistent data, and rising costs.

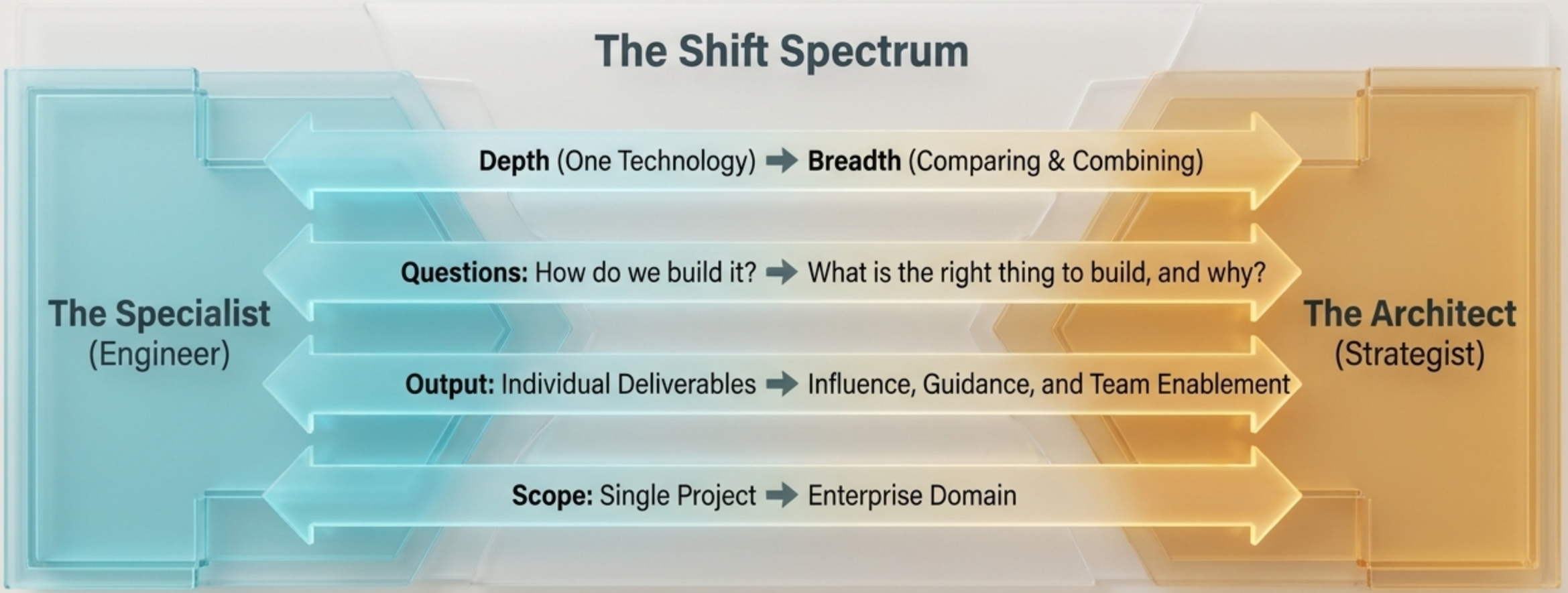
Core Maxim: Good architecture makes change cheap; poor architecture makes every change expensive.



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The Architect’s Evolution

The Shift Spectrum

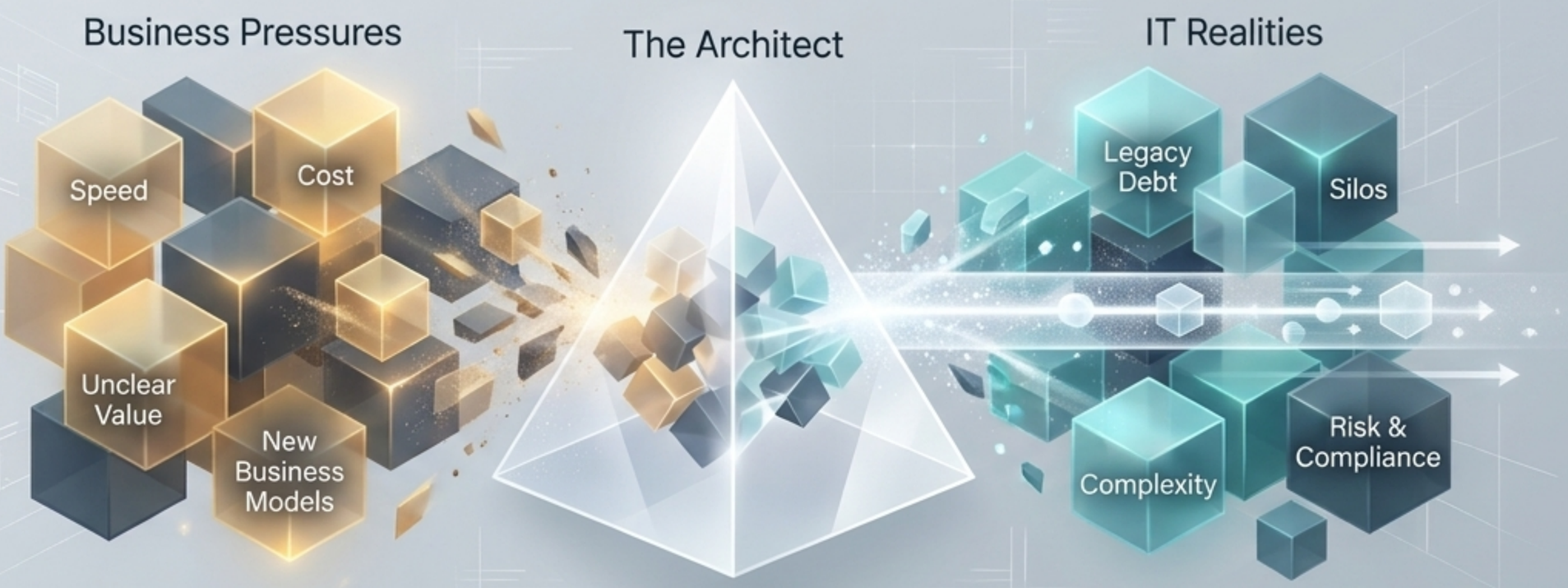


The ultimate self-check: Can you explain a technical decision to the CFO in terms of cost, risk, and benefit?

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Bridging the Divide: The Architect as Translator

The Collision Diagram



Architects work exactly where business pressure and technical reality collide. Their role is pure translation: turning business goals into technical direction, and technical constraints into business consequences.

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One System, Five Perspectives

Business Lens

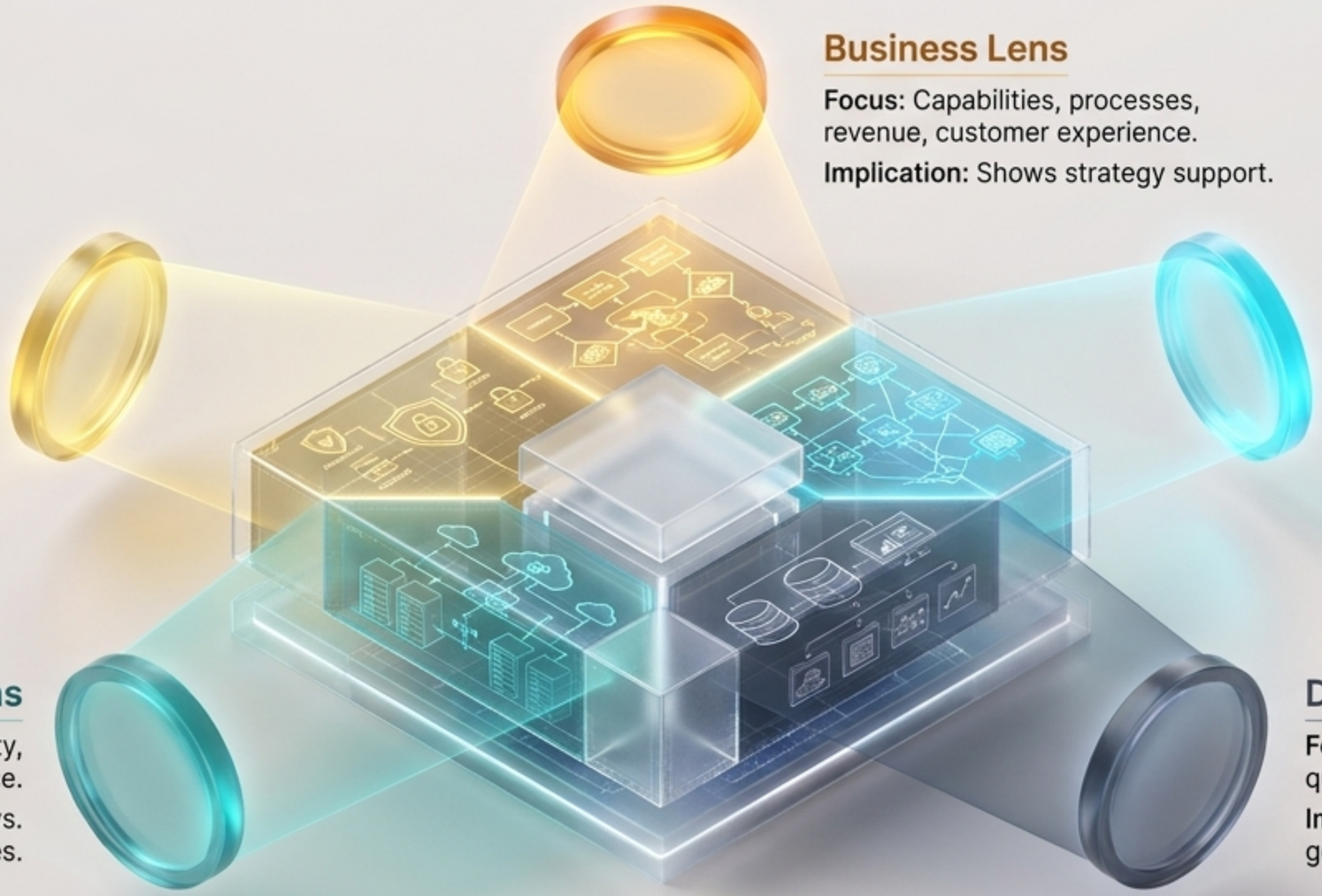
Focus: Capabilities, processes, revenue, customer experience.

Implication: Shows strategy support.

Security Lens

Focus: Threats, controls, compliance.

Implication: Must be designed in, not added later.



Application Lens

Focus: Functionality, integration, lifecycle.

Implication: Drives build vs. buy decisions.

Infrastructure Lens

Focus: Hosting, capacity, resilience.

Implication: Cloud vs. on-prem choices.

Data Lens

Focus: Ownership, privacy, quality, analytics.

Implication: Drives data governance.

A diagram that satisfies engineers will usually confuse executives. The architect must move fluidly between views.

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The Architect Role Taxonomy

Role	Scope	Primary Focus	Key Implication / Driver
Enterprise	Whole Organization	IT/Strategy Alignment	Roadmaps & Standards
Business	Enterprise	Capabilities & Value Streams	Strategy to Execution
Solution	Specific Initiative	End-to-End Design	Fitting the wider landscape
Application	Domain/Project	Structure & Integration	Application Lifecycle
Data	Enterprise/Domain	Modeling, Sharing, Analytics	Data as a shared asset
Infrastructure	Enterprise/Domain	Platforms, Networks, Cloud	Hosting & Capacity
Security	Cross-cutting	Threat Models & Controls	Compliance & Risk Mitigation

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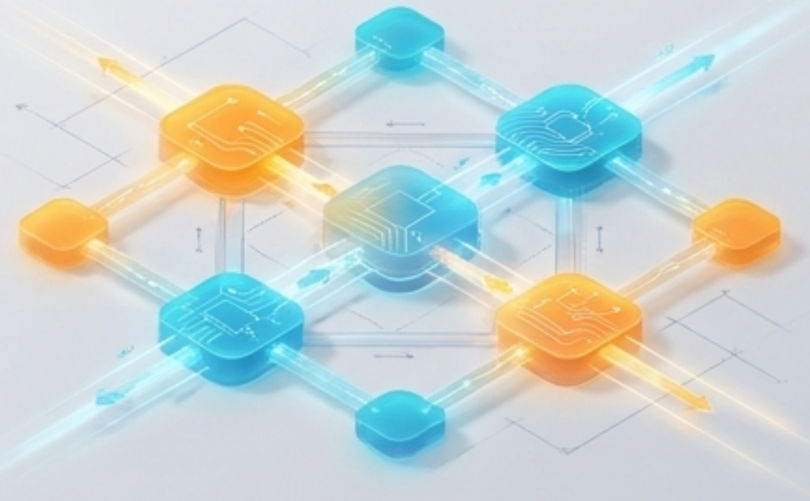
The New Enterprise Paradigm

Traditional Enterprise



- Hierarchical & Process-driven
- Long Planning Cycles
- Siloed Departments
- Build-it-all Internally
- Physical / On-Prem

The New Paradigm

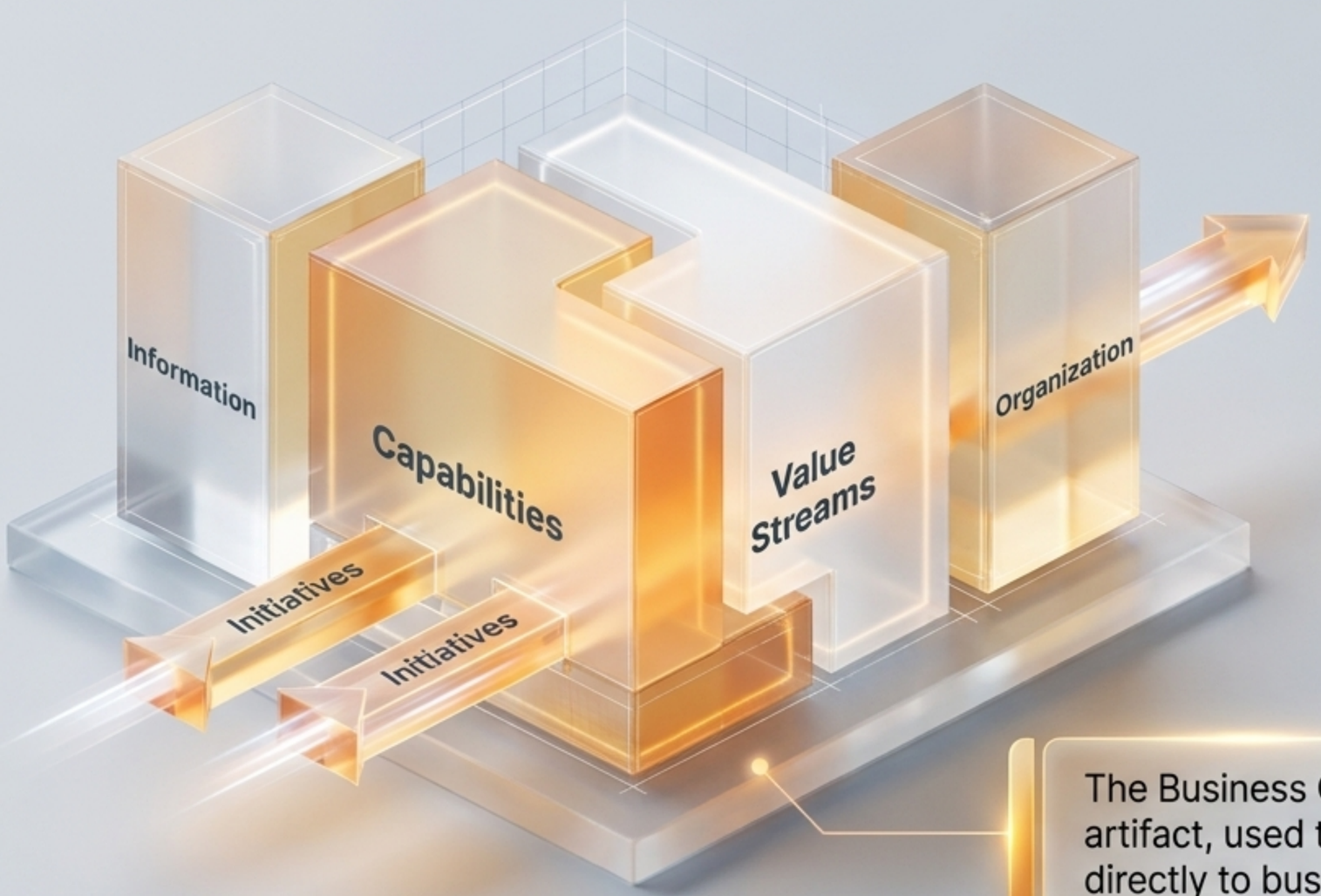


- Customer-centric (Journey-based)
- Agile, Adaptive, Iterative
- Cross-functional, Product-aligned Teams
- Platform & Ecosystem (APIs, Partners)
- Cloud & Service-based

Implication: Architecture must be lighter and more continuous. Set guardrails and reusable building blocks instead of massive upfront blueprints.

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The 5 Pillars of Business Architecture



1. **Capabilities:** What the business can do (Stable; the what, not the how).
2. **Value Streams:** The end-to-end sequence delivering customer value (e.g., Quote to Policy Issue).
3. **Organization:** Structure, roles, and partners executing the work.
4. **Information:** Key business concepts and data relationships.
5. **Initiatives:** Projects and programs driving change.

The Business Capability Map is the central executive artifact, used to prioritize investment and link IT systems directly to business needs.

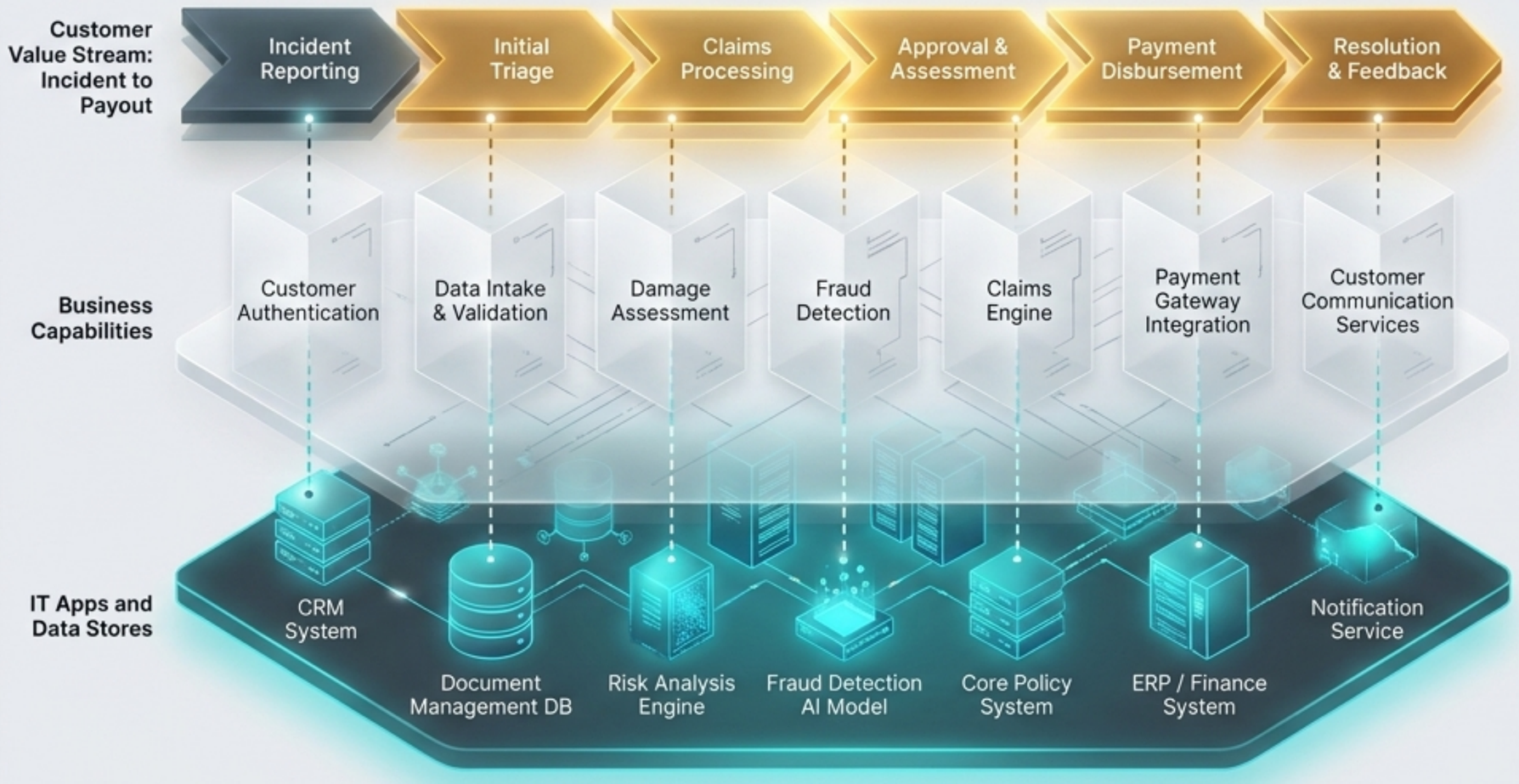
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Synthesis: Value Stream Integration

Connecting the Customer to the Code.

Mapping capabilities to value stages reveals exactly which IT systems are mission-critical, where fatal gaps exist, and where value is actually generated.

A slow approval stage lacking IT support immediately becomes a clear investment target.

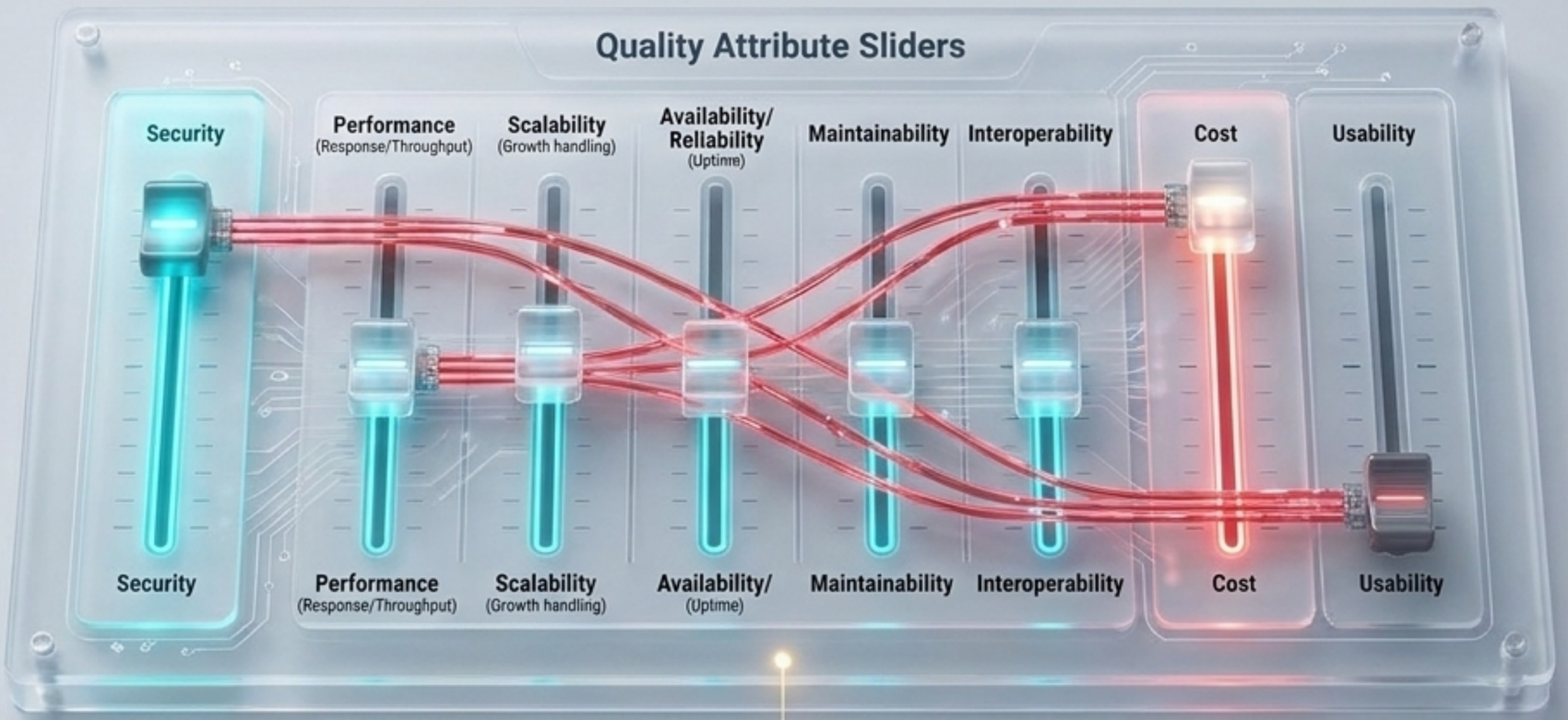


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The Science of Design: Quality Attributes

Quality attributes dictate how well a system works, not what it does.

They frequently conflict.

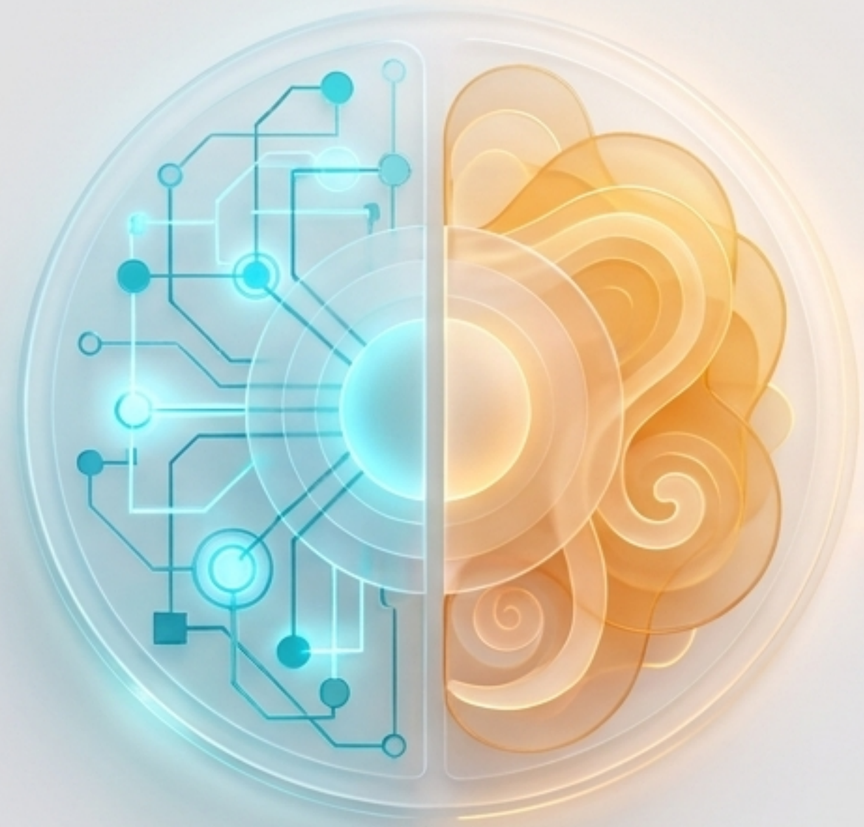


The Architect's Mandate: The primary job in design is to clarify which attributes matter most, make the trade-offs explicit, and document them.

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The Art of Architecture: Human Dynamics

Technical Design
(Science)



Human Dynamics
(Art)

Influence Without Authority:
Persuading through credibility and evidence, not command.

Engagement Posture:
Moving fluidly between Advisor (guidance), Partner (co-design), and Gatekeeper (risk control).

Emotional Awareness:
Detecting resistance, fear of change, and politics early.

Technical correctness does not guarantee adoption. An architecture that people do not understand or trust will not be followed.

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Demystifying Business Requirement Architecture

Requirements are an architecture. They have structure, layers, and relationships.

Business Context & Stakeholders

Actor Goals & Use Cases

Detailed Scenarios, Rules & Traceability



Core Principles

Start from Goals

Every requirement must trace back to a business objective.

Model, Don't List

Use structured models to show system interaction.

Progressive Refinement

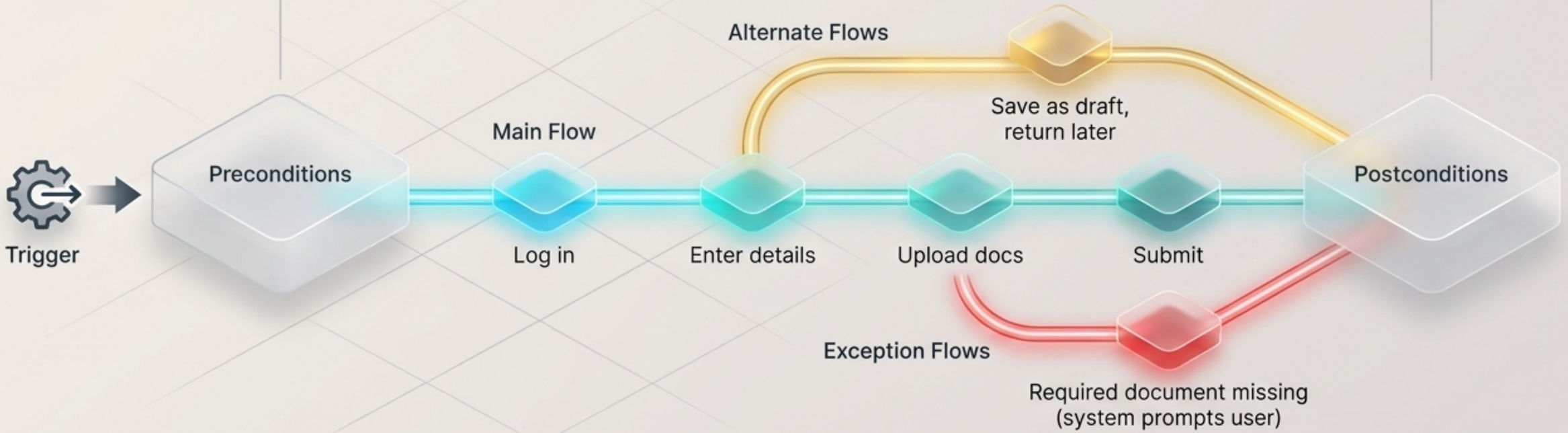
Add detail only as you move down the funnel, reducing rework.

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Anatomy of a Use Case Scenario

Example: Insurance Claim

The Use Case Storyboard



Good scenarios read like a short story with a clear beginning, middle, and end.

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Use Case Modeling: Pitfalls & Patterns

The Pitfall	The Architectural Fix
Too much technical detail (screens/code)	Describe business intent only; leave design for later.
Wrong granularity (“Manage everything”)	Aim for a single goal completed in one sitting.
Missing exception flows (Happy path only)	For every single step, forcefully ask “what could go wrong?”
Hidden business rules buried in text	Extract and catalog in a centralized rules pattern.

Leveraging Patterns

Reuse proven architectural requirement patterns to ensure consistency:

- Goal-first pattern
- CRUD pattern (Create, Read, Update, Delete)
- Approval pattern

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Closing the Loop: Governance & ROI



Governance is how architectural direction is enforced and exceptions are handled. Good governance is proportionate—enabling delivery rather than blocking it.

Measure success in the language of the business: avoided costs, reduced time-to-market, mitigated risk, and increased organizational agility.

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